

JKBOSE 2026

CLASS 11TH

Photosynthesis in Higher Plants

LECTURE-05

BOTANY

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Topics To Covered

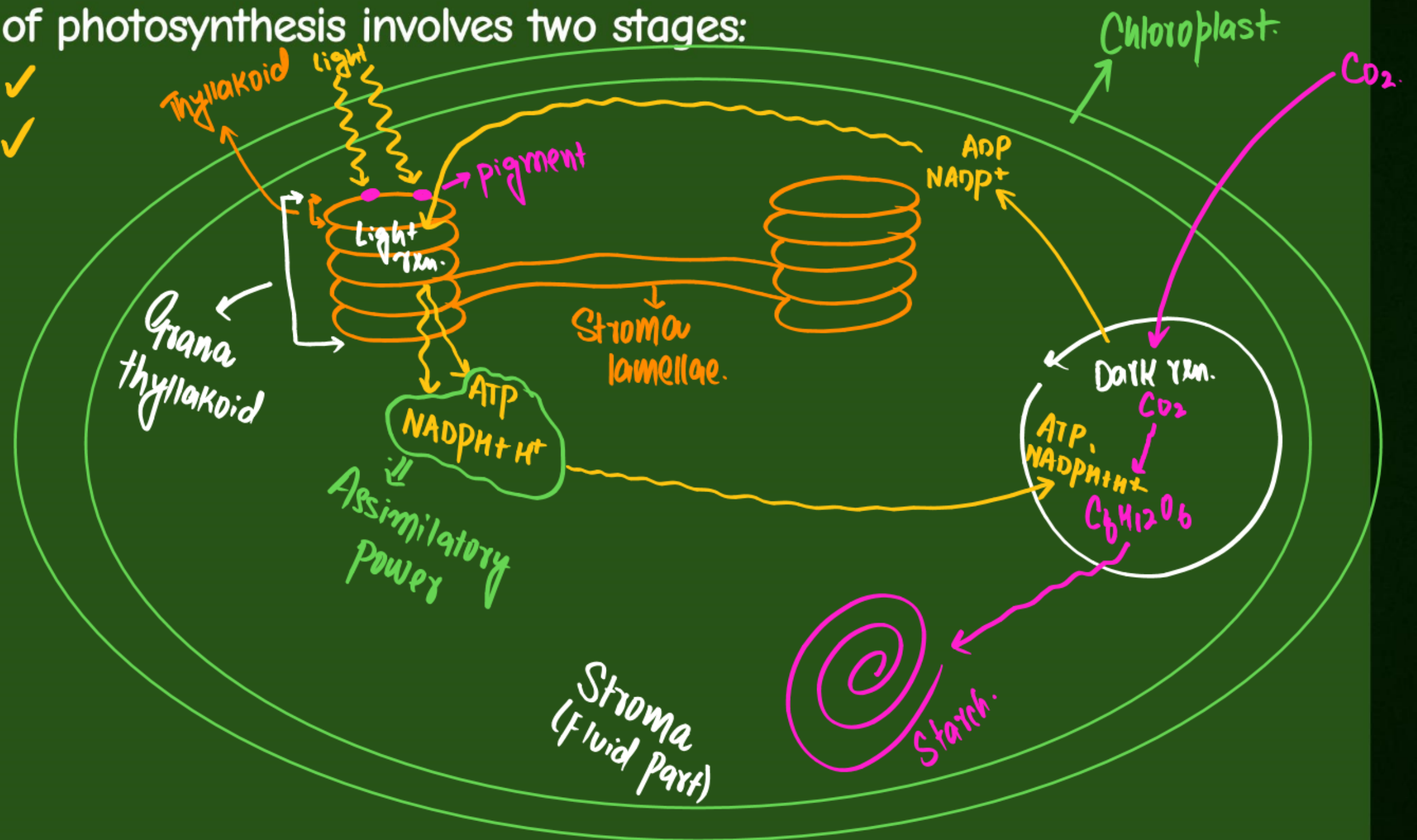
1 Mechanism of Photosynthesis

2 ...

Mechanism of Photosynthesis

→ Complex process of photosynthesis involves two stages:

- a) Light reaction ✓
- b) Dark Reaction ✓



Light Reaction

- AKA photochemical reaction.
- Site: Thylakoid membrane.
- Purpose: Formation of Assimilatory power i.e, ATP and NADPH+H⁺
- It directly depends on light.

Dark Reaction

- AKA Biosynthetic phase because Glucose is formed.

• Site: Stroma of Chloroplast.

• Purpose: Formation of Glucose by using ATP and NADPH+H⁺.

• It depends upon products of light reaction.

Note: The name "Dark reaction" to Biosynthetic phase is a misnomer because it does not require darkness to occur.

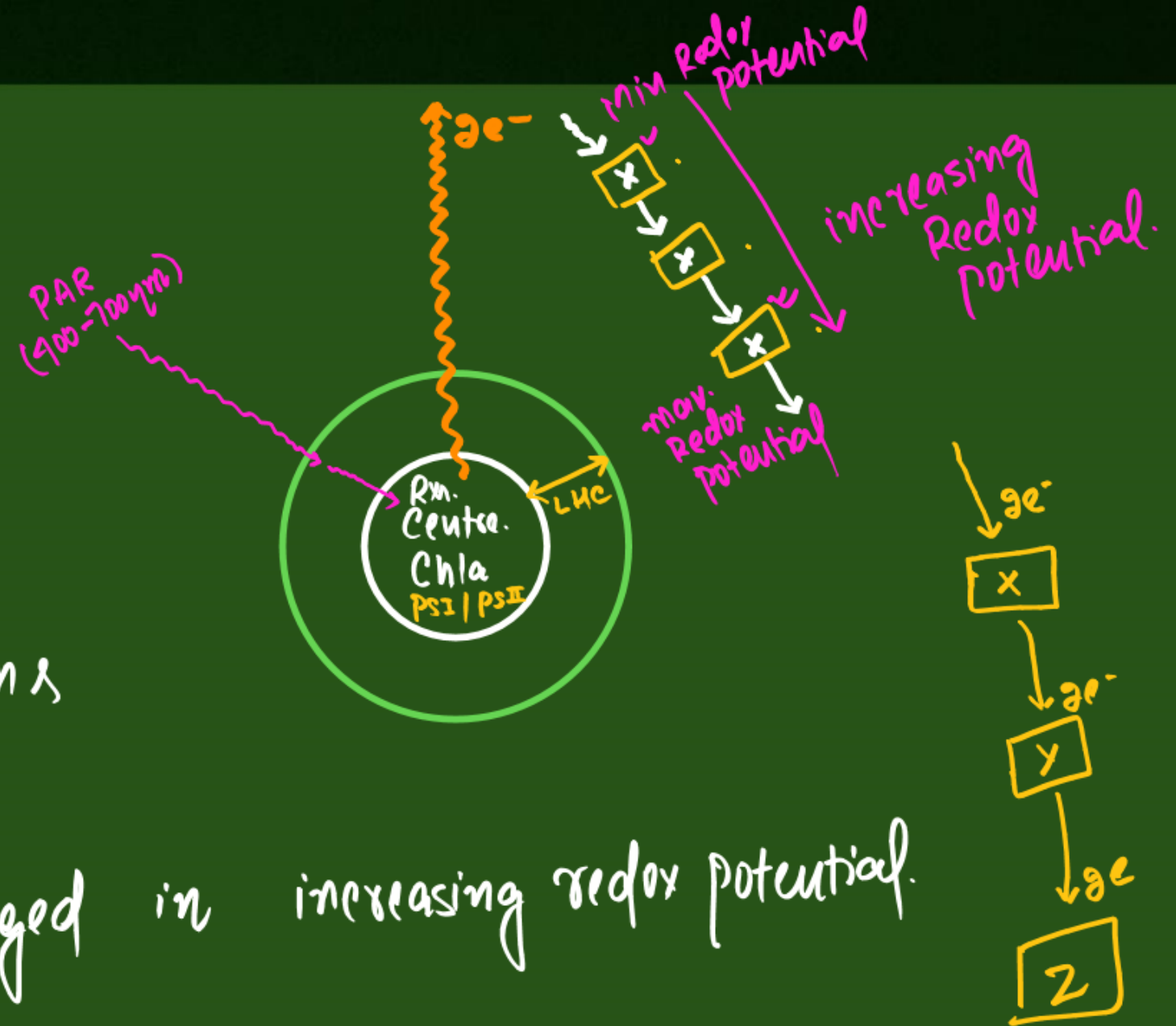
→ It requires products of light rxn. i.e, ATP & NADPH+H⁺ to occur.

Light Reaction

- It involves Electron transport.
- The electrons excited from PSI and PS-II are carried by electron transporters in an electron transport chain.
- This electron transport chain contains Cytochrome Complex.

Note! The e^- transporters/carriers are arranged in increasing redox potential.

∴ During light reaction - photophosphorylation occurs.



Photophosphorylation

Photo → light
phosphorylation → formation of ATP.

≡ Photophosphorylation → It is formation of ATP by using light energy.

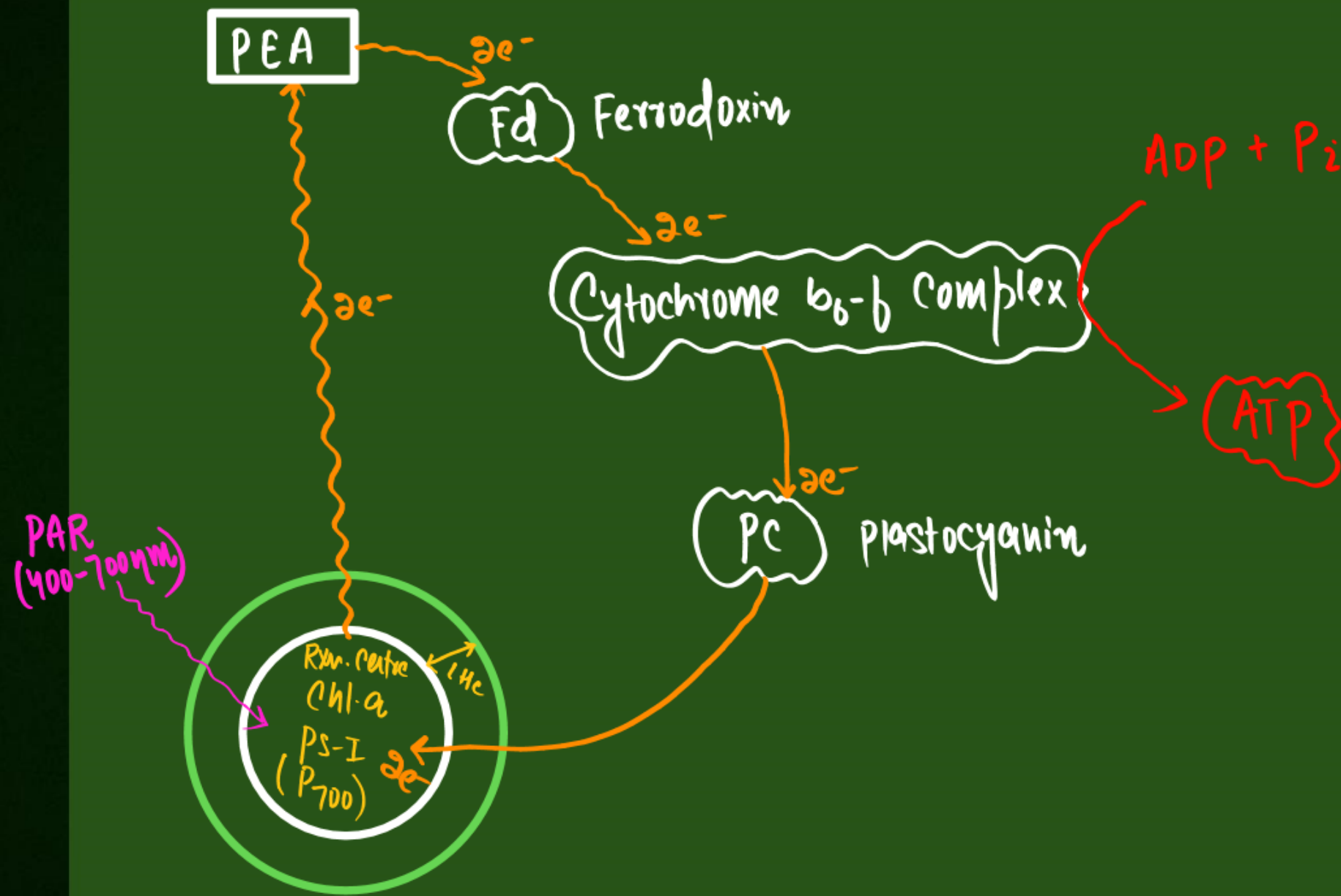


≡ Photophosphorylation — is of two types:

a. Cyclic photophosphorylation.

b. Non-cyclic photophosphorylation.

a. Cyclic Photophosphorylation



≡ Here only PS-I is involved.

≡ Only ATP is formed.

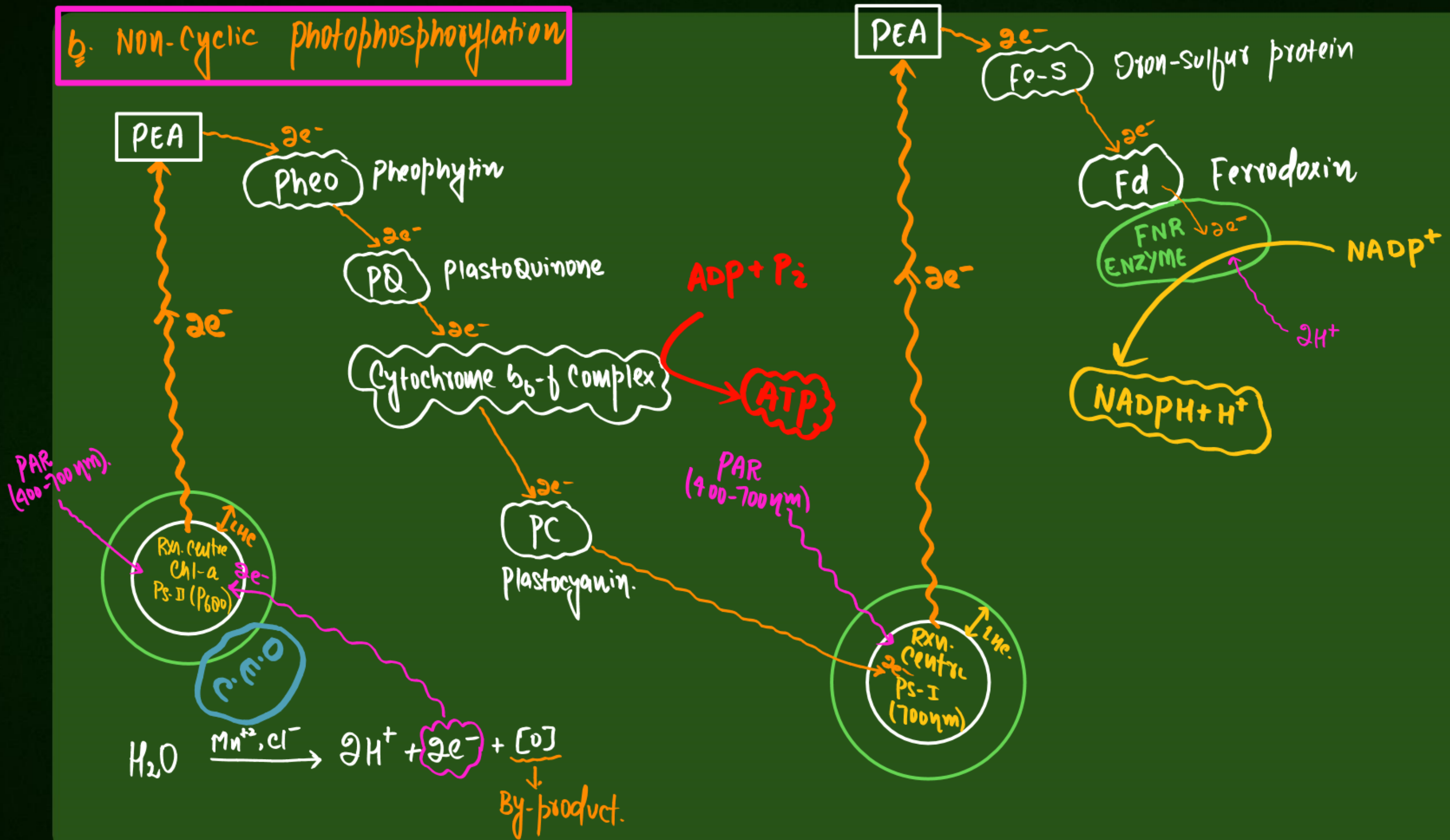
≡ Electron comes back to reaction centre of PS-I. i.e., why the name cyclic photophosphorylation.

≡ Site: Stroma lamellae.

PS-II is absent.

It also lacks FNR (Ferredoxin NADP⁺ reductase enzyme).

b. Non-cyclic photophosphorylation



- ≡ Here both PS-I and PS-II are involved.
 - ≡ e^- exited from PS-II $\rightarrow$ does not come back to reaction centre of PS-II
i.e, why the name non-cyclic photophosphorylation.
 - ≡ Electron follows zig-zag path i.e, why it is also called as "Z-scheme"
 - ≡ Both ATP and $\text{NADPH} + \text{H}^+$ are formed.
 - ≡ Site: Grana thylakoid.
- In non-cyclic photophosphorylation — O_2 is evolved due to splitting of H_2O
 $\downarrow$
Photolysis of H_2O .

Chemosmotic Hypothesis

Next lecture ✓✓

QUESTIONS

Which of the following is not a product of light reaction of photosynthesis?
[NEET-2018]

A ATP ✓

B NADH

C Oxygen ✓

D NADPH ✓

QUESTIONS

In light reaction, plastoquinone facilitates the transfer of electrons from:

(2020)

- A** Cytb₆f complex to PS-I
- B** PS-I to NADP⁺
- C** PS-I to ATP synthase
- D** PS-II to Cytb₆f complex

QUESTIONS

During non-cyclic photophosphorylation, when electrons are lost from the reaction centre at PS II, what is the source which replaces these electrons?

(2020-Covid)

- A** Water
- B** Carbon dioxide
- C** Light
- D** Oxygen

QUESTIONS

Which one of the following concerns photophosphorylation?

(2003)



QUESTIONS

In photosynthesis, energy from light reaction to dark reaction is transferred in the form of: (2002)

- ☐ A ADP
- ☒ B ATP
- ☐ C RuBP
- ☐ D Chlorophyll

HOMEWORK

Revision





Thank
You